

C-GLASS FLOW

Set 'run-opts' via 'parse-opts'

↓
Set 'SimulationManager' via 'InitManager'

- ↳ Accepts 'run-opts' as input
- ↳ Parses 'run-opts' then 'InitLogger'
- ↳ Check AppendParams
 - ↳ looks for recursive YAML files
- ↳ Load Default Params
 - ↳ Sets initialized params to default
- ↳ Check Random Params
 - ↳ If input has a range, set within it
- ↳ Count Variations
 - ↳ If input includes sequences, calc. tot. #
- ↳ Generate Parameters
 - ↳ Make combinatorics of param files

* Same obj

Call 'RunManager' for 'SimulationManager'

- ↳ Write Params
 - ↳ convert node combos to YAML files
- ↳ 'return' if run-opts . blank flag
- ↳ ProcessOutputs if analysis flag
 - * ↳ call ProcessOutputs for all Sims.
- ↳ Run Simulations otherwise
 - * ↳ call 'Run' for all sims

* 'new'
↓
call f(x)
↓
delete

↳ Turn page for Flow chart
(ProcessOutputs TBD)

C-GLASS::SIMULATION::RUN

Accepts XAML Node of params as input

Outputs sim params via XAML::Emitter

Call 'Init' for 'ParamsParser'

- ↳ Accepts XAML node as input
- ↳ Parses XAML node on a per-species basis
- ↳ checks for species label duplicates

Call 'InitSimulation' for 'Simulation'

- ↳ Call 'ParseSystemParameters' for 'ParamsParser'
- ↳ converts node to sys-params struct

INIT
ALL
OBJECTS

- ↳ Set 'run-name'

- ↳ initialize new RNG w/ seed

Call 'RunSimulation' * flow chart on next page

- ↳ main KMC-BD loop

- ↳ integrates, writes output, draws

- ↳ Requires its own flowchart

Call 'ClearSimulation'

- ↳ Call 'close' for 'output_mgr'

- ↳ * Call 'ClearSpecies'

- ↳ Call 'Cleanup' for all species

- ↳ 'delete' all species, class-specific

- ↳ * Call 'Clear' for 'ix_mgr'

- ↳ * Call 'Clear' for 'graphics'

- ↳ 'delete' rng

- ↳ 'delete' cortex

- ↳ Call Tracker::PrintTrack

* Same flow
* Collapse all
then delete

Done!

RUNSIMULATION() Flow

Set 'time-', 'i-step' and 'delta-diff' to 0
↓
Run main KMC-BD 'FOR' loop

- ↳ Increment 'time-' and 'i-step-'; Check Midstep
 - ↳ $\text{time-} \leftarrow \text{step-} \times \text{Obj::GetDelta}$
 - ↳ if dynamic timestep, round for i-step-
- ↳ PrintComplete() outputs sim progress
- ↳ Zero Forces() zeroes out all force for all species
- ↳ Interact() calls ix_mgr-.Interact()

all within InteractionManager

- ↳ 'return' if no interactions or boundaries
- ↳ CheckUpdateObjects checks if any interactors/lists need to be updated
- ↳ Xlink-.UpdateCrosslinks updates all Xlink positions/kinetic rates, checks knockout/etc.
- ↳ CheckUpdateXlinks checks if any xlink interactors need to be updated.
- ↳ CheckUpdateInteractors " for pair-wise
- ↳ *Calculate Interactions
 - ↳ Clear object interactions
 - ↳ *same flow Calculate Pair Interactions
 - ↳ Calculate Boundary Interactions
- ↳ Check for excess force, 'return' if so
 - ↳ also flags decrease dynamic timestep = T
- ↳ *Apply Interactions for Pair and Boundary
- ↳ increment i-update by 1

- ↳ Adjust dynamic delta if flag is set
- ↳ WriteOutput() writes all data to files
- ↳ Integrate() calls 'updatePositions' for all species
- ↳ Statistics() calculates thermodynamic quantities
- ↳ Draw() draws graphics if enabled
- ↳ check for early exit

Done!